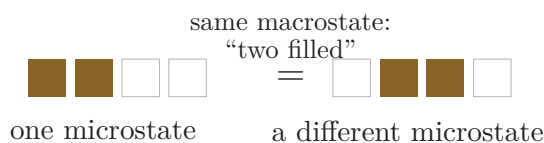


How Many Ways Can It Look the Same?

Name: _____

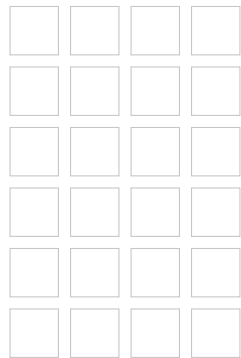
A count of everything that could be true and still look exactly like what's visible from outside, built from coins, cubes, and marbles before a single joule of heat enters the picture.



2. Four coloured cubes

listing starts to strain

Four cubes sit in a row; each is brass or plain, and only the colour sequence matters. List every distinct row with exactly two brass cubes.



How many rows did you find? _____ Are you confident you found all of them? What would make you more confident?

3. Six students around a table

the count grows

Four blue shirts, two red. Without listing every seating pattern, estimate how many more arrangements this has than the four-cube case, and explain your reasoning.

4. Marbles in boxes

which split wins

Four identical marbles split between two boxes. Complete the table using $\binom{4}{k}$ for each split.

Split (4–0, 3–1, ...)	k	$\Omega = \binom{4}{k}$
4–0	0	
3–1	1	
2–2	2	
1–3	3	
0–4	4	

Which split corresponds to the largest number of microscopic arrangements? _____

5. Eight particles

when listing gives up

Use Pascal's triangle rather than listing to fill in $\Omega(8, k)$ for every k .

$$\begin{array}{cccccccccc}
 & & & & & & & & & & \\
 & & & & & & & & & & \binom{0}{0} \\
 & & & & & & & & & & \binom{1}{0} \quad \binom{1}{1} \\
 & & & & & & & & & & \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \\
 & & & & & & & & & & \binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \\
 & & & & & & & & & & \binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4} \\
 & & & & & & & & & & \binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5} \\
 & & & & & & & & & & \binom{6}{0} \quad \binom{6}{1} \quad \binom{6}{2} \quad \binom{6}{3} \quad \binom{6}{4} \quad \binom{6}{5} \quad \binom{6}{6} \\
 & & & & & & & & & & \binom{7}{0} \quad \binom{7}{1} \quad \binom{7}{2} \quad \binom{7}{3} \quad \binom{7}{4} \quad \binom{7}{5} \quad \binom{7}{6} \quad \binom{7}{7} \\
 & & & & & & & & & & \binom{8}{0} \quad \binom{8}{1} \quad \binom{8}{2} \quad \binom{8}{3} \quad \binom{8}{4} \quad \binom{8}{5} \quad \binom{8}{6} \quad \binom{8}{7} \quad \binom{8}{8}
 \end{array}$$

$\Omega(8, k)$ for $k = 0, \dots, 8$:

What happened to the strategy of listing every arrangement by hand?

6. Only now, probability

most likely macrostates

Assume every individual microstate is equally likely. The total number of microstates for N binary items is 2^N .

$$P(k) = \frac{\Omega(N, k)}{2^N}$$

For $N = 8$, compute $P(4)$ and $P(0)$.

$$P(4) = \text{_____} \quad P(0) = \text{_____}$$

Did counting or probability come first in this investigation? Explain why that order matters.

7. Introduce Ω

name what you have been computing

$$\Omega(N, k) = \frac{N!}{k!(N-k)!}$$

Verify this formula reproduces one of your Pascal's-triangle values above.

8. Why logarithm?

two systems combine Sys-

tem A has $\Omega_A = 20$ microstates; system B has $\Omega_B = 50$. Combined:

$$\Omega_A \times \Omega_B = \underline{\hspace{2cm}}$$

Show that $\ln(\Omega_A \times \Omega_B) = \ln \Omega_A + \ln \Omega_B$ using your calculator.

$$\ln \Omega_A = \underline{\hspace{2cm}} \quad \ln \Omega_B = \underline{\hspace{2cm}} \quad \text{sum} = \underline{\hspace{2cm}} \quad \ln(\Omega_A \Omega_B) = \underline{\hspace{2cm}}$$

A second question about the same Ω : how many yes/no questions would it take to identify one arrangement among Ω equally likely ones?

$$2^q = \Omega \quad \implies \quad q = \log_2 \Omega$$

Compute q for $\Omega_A = 20$: $\underline{\hspace{2cm}}$

9. Only now, heat

Boltzmann's equation

$$S = k_B \ln \Omega$$

Adding heat raises a system's total energy. Explain, in your own words, why more energy on hand should mean a larger Ω .

Why is this equation better understood as a restatement of the counting exercise than as a new definition?

10. A stronger ordering

compare the two hierarchies Fill

in the missing cell, then compare the two orderings directly.

The usual order	The order used here
Thermodynamics	Counting
Statistical mechanics	Probability
Information	Information
	Thermodynamics

Which order was assumed least and earned most? Justify your answer with reference to at least two specific stages above.

11. Field notes

talk generates; writing stabilises

Before discussion:

I first assumed that...

One quantity I can calculate is...

After discussion:

My current conjecture about what entropy is counting is...

The part I still cannot connect is...

12. Core results

what this enquiry actually earned

The relations established along the way.

$$\Omega(N, k) = \frac{N!}{k!(N-k)!}$$

$$\ln(\Omega_A \Omega_B) = \ln \Omega_A + \ln \Omega_B$$

$$q = \log_2 \Omega$$

$$S = k_B \ln \Omega$$

Concepts earned, in order.

- Microstate and macrostate were distinguished before either word was supplied.
- Listing gives way to a formula precisely when listing stops being a reasonable strategy, not before.
- Probability turns out to ride on counting that had already been done.
- The logarithm is the operation that turns multiplying into adding, and separately, that measures information in yes/no questions.
- Heat raising entropy is a statement about how many arrangements become newly accessible, not a definition to memorise.

Every substantial claim should be accompanied by a reason, calculation, diagram, table, or counterexample.